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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 26th August Morning Shift

JEE Main Question Paper with Answer Key

26 Aug 2021	90	20	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	22 questions
Mathematics	20 questions
Physics	28 questions
Out of syllabus	20 questions
Answer key	at the end

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Chemistry 22 questions

Q1. MCQ (Single Correct Answer)

Which one of the following complexes is violet in colour?

(A) $[Fe(CN)_6]^{4-}$

(B) $[Fe(SCN)_6]^{4-}$

(C) $Fe_4[Fe(CN_6)]_3 \cdot H_2O$

(D) $[Fe(CN)_5NOS]^{4-}$

Open solution

Q2. MCQ (Single Correct Answer)

Which one of the following when dissolved in water gives coloured solution in nitrogen atmosphere?

(A) $CuCl_2$

(B) AgCl

(C) $ZnCl_2$

(D) Cu_2Cl_2

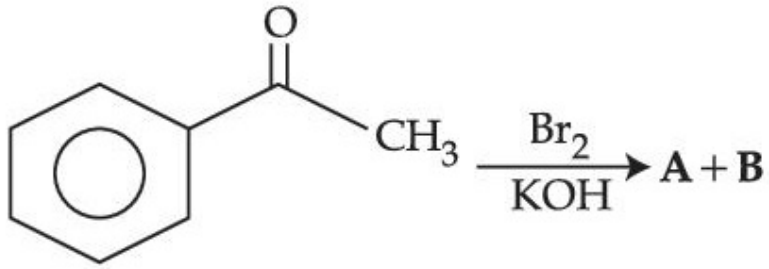
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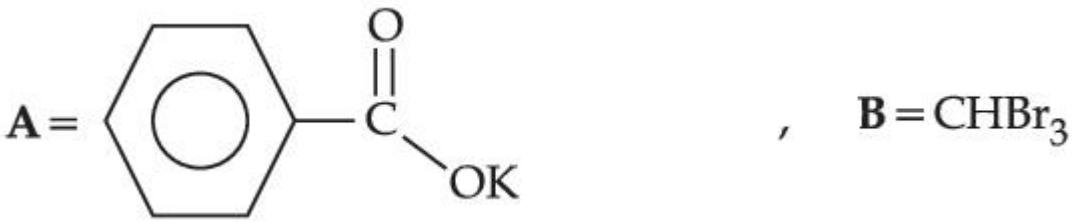
Q3. MCQ (Single Correct Answer)

Open solution

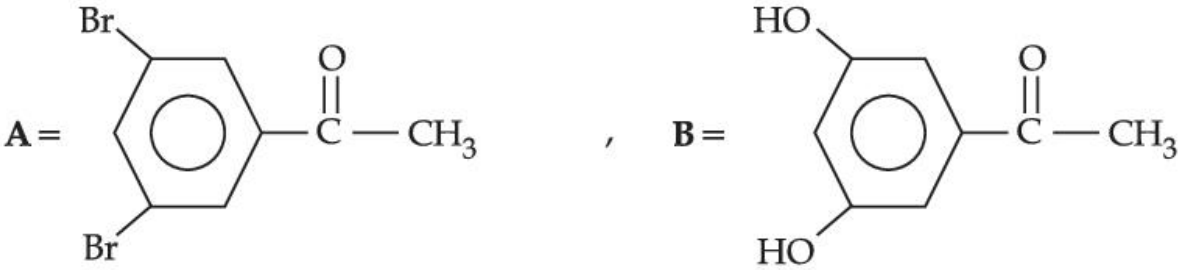
The major products formed in the following reaction sequence A and B are :



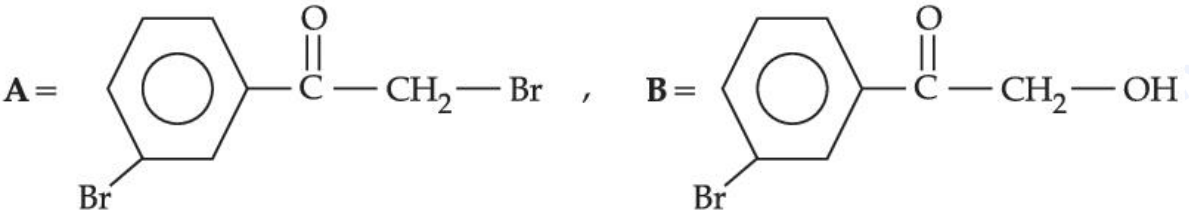
(A)



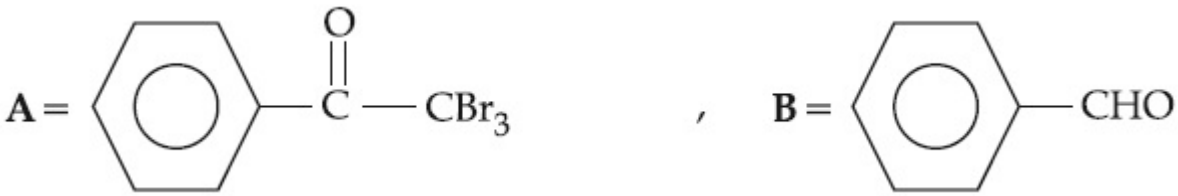
(B)



(C)



(D)



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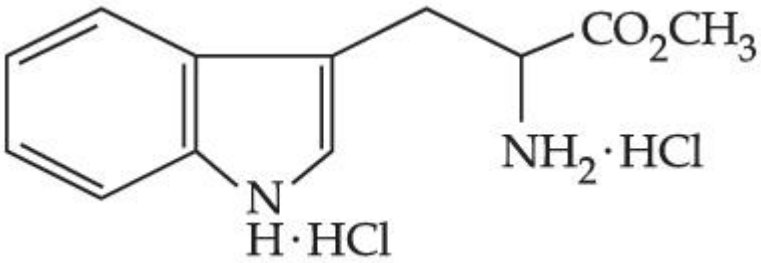
Q4. MCQ (Single Correct Answer)

Open solution

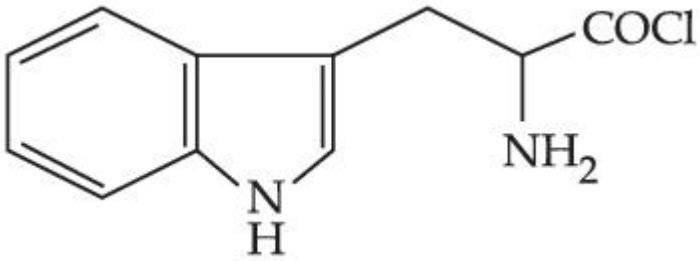
The major product formed in the following reaction is :



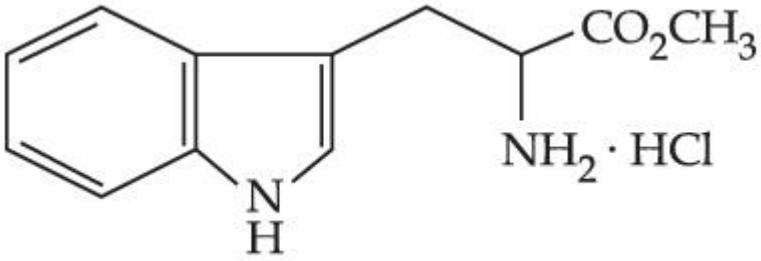
(A)



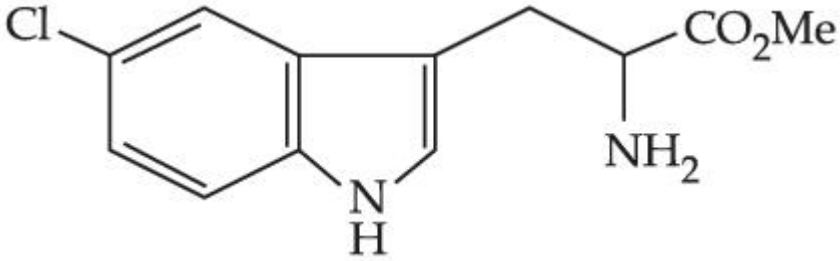
(B)



(C)



(D)

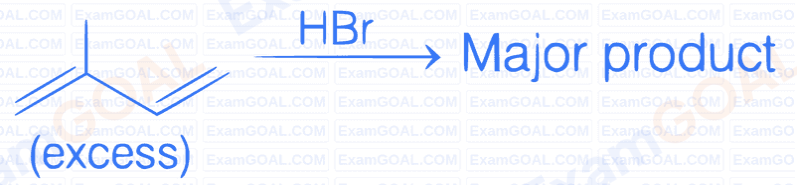


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Q5. MCQ (Single Correct Answer)

Open solution

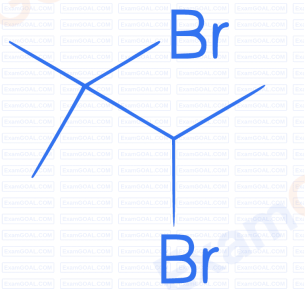
The major product formed in the following reaction is :



(A)



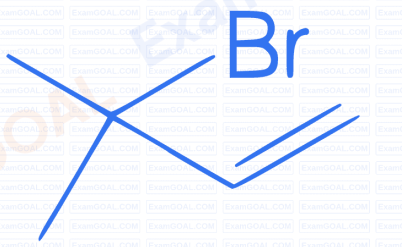
(B)



(C)



(D)



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Q6. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : The limiting molar conductivity of KCl (strong electrolyte) is higher compared to that of CH_3COOH (weak electrolyte).

Statement II : Molar conductivity decreases with decrease in concentration of electrolyte.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is true but Statement II is false.

(B) Statement I is false but Statement II is true.

(C) Both Statement I and Statement II are true.

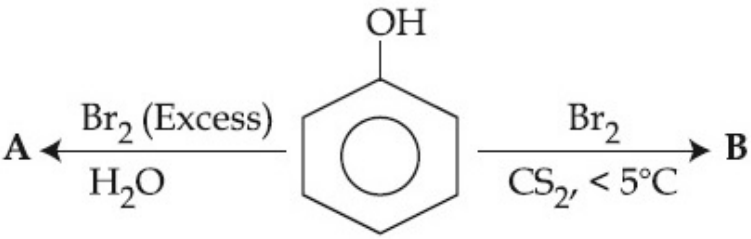
(D) Both Statement I and Statement II are false.

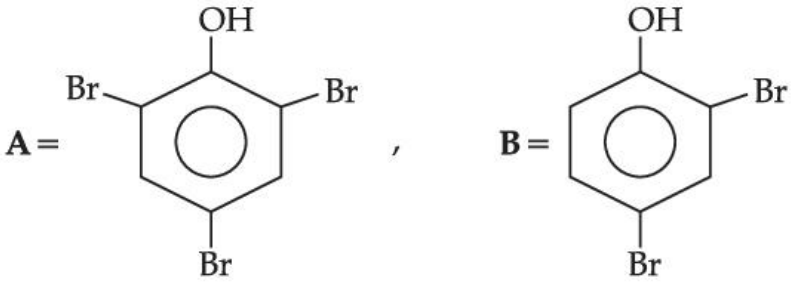
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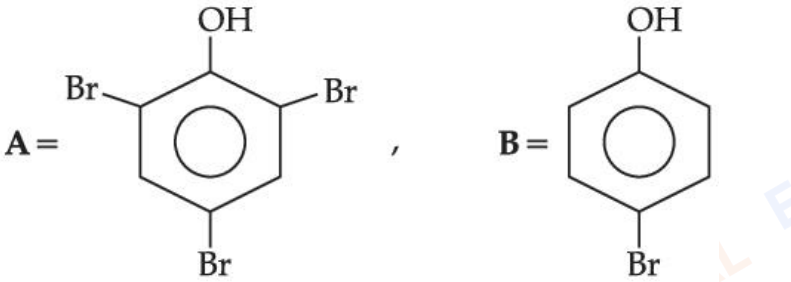
Q7. MCQ (Single Correct Answer)

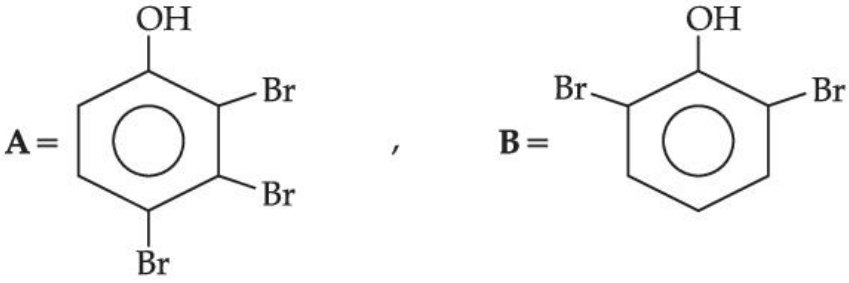
Open solution

The correct options for the products **A** and **B** of the following reactions are :



(A) 

(B) 

(C) 

(D) 

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Q8. MCQ (Single Correct Answer)

Open solution

Given below are two statements.

Statement I : In the titration between strong acid and weak base methyl orange is suitable as an indicator.

Statement II : For titration of acetic acid with NaOH phenolphthalein is not a suitable indicator.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Statement I is true but Statement II is false

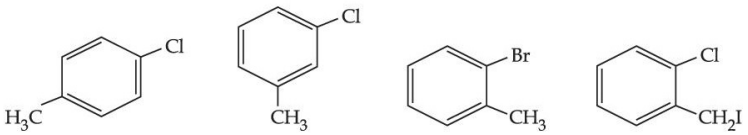
(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q9. MCQ (Single Correct Answer)

Open solution

Among the following compounds I-IV, which one forms a yellow precipitate on reacting sequentially with (i) NaOH (ii) dil. HNO_3 (iii) $AgNO_3$?



I II III IV

(A) II (B) IV (C) I (D) III

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Q10. MCQ (Single Correct Answer)

Open solution

Given below are two statements.

Statement I : According to Bohr's model of an atom, qualitatively the magnitude of velocity of electron increases with decrease in positive charges on the nucleus as there is no strong hold on the electron by the nucleus.

Statement II : According to Bohr's model of an atom, qualitatively the magnitude of velocity of electron increases with decrease in principal quantum number.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Q11. MCQ (Single Correct Answer)

Open solution

The correct sequential addition of reagents in the preparation of 3-nitrobenzoic acid from benzene is :

(A) $Br_2/AlBr_3$, HNO_3/H_2SO_4, Mg/ether, CO_2, H_3O^+

(B) $Br_2/AlBr^3$, NaCN, H_3O^+, HNO_3/H_2SO_4

(C) $Br_2/AlBr_3$, HNO_3/H_2SO_4, NaCN, H_3O^+

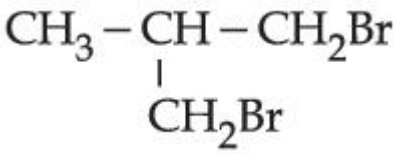
(D) HNO_3/H_2SO_4, $Br_2/AlBr_3$, Mg/ether, CO_2, H_3O^+

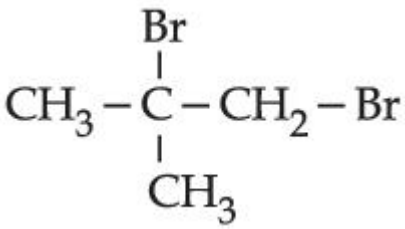
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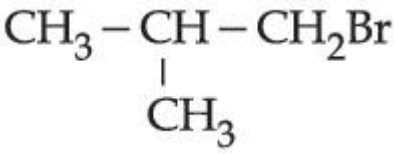
Q12. MCQ (Single Correct Answer)

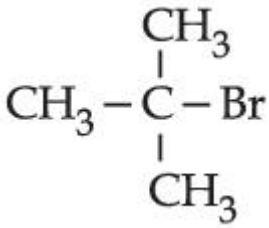
Excess of isobutane on reaction with Br_2 in presence of light at 125°C gives which one of the following, as the major product?

Open solution

(A) 

(B) 

(C) 

(D) 

Q13. Numerical

AB_3 is an interhalogen T-shaped molecule. The number of lone pairs of electrons on A is _____. (Integer answer)

Open solution

Q14. Numerical

These are physical properties of an element

(A) Sublimation enthalpy

(B) Ionisation enthalpy

(C) Hydration enthalpy

(D) Electron gain enthalpy

The total number of above properties that affect the reduction potential is _____ (Integer answer)

Open solution

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Q15. Numerical

Of the following four aqueous solutions, total number of those solutions whose freezing point is lower than that of 0.10 M C_2H_5OH is _____ (Integer answer)

(i) 0.10 M $Ba_3(PO_4)_2$

(ii) 0.10 M Na_2SO_4

(iii) 0.10 M KCl

(iv) 0.10 M Li_3PO_4

Open solution

Q16. Numerical

The OH^- concentration in a mixture of 5.0 mL of 0.0504 M NH_4Cl and 2 mL of 0.0210 M NH_3 solution is $x \times 10^{-6}$ M. The value of x is _____. (Nearest integer)

[Given $K_w = 1 \times 10^{-14}$ and $K_b = 1.8 \times 10^{-5}$]

Open solution

Q17. Numerical

The number of 4f electrons in the ground state electronic configuration of Gd^{2+} is _____.
[Atomic number of Gd = 64]

Open solution

Q18. Numerical

The ratio of number of water molecules in Mohr's salt and potash alum is _____ $\times 10^{-1}$.
(Integer answer)

Open solution

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Q19. Numerical

The following data was obtained for chemical reaction given below at 975 K.

$2NO_{(g)} + 2H_{2(g)} \rightarrow N_{2(g)} + 2H_2O_{(g)}$

	$[NO]$ mol L^{-1}	$[H_2]$ mol L^{-1}	Rate mol $L^{-1}s^{-1}$
(A)	8×10^{-5}	8×10^{-5}	7×10^{-9}
(B)	24×10^{-5}	8×10^{-5}	2.1×10^{-8}
(C)	24×10^{-5}	32×10^{-5}	8.4×10^{-8}

The order of the reaction with respect to NO is _____. [Integer answer]

Open solution

Q20. Numerical

The Born-Haber cycle for KCl is evaluated with the following data :

$\Delta_f H^\Theta$ for KCl = -436.7 kJ mol^{-1} ;

$\Delta_{sub} H^\Theta$ for K = 89.2 kJ mol^{-1} ;

$\Delta_{ionization} H^\Theta$ for K = 419.0 kJ mol^{-1} ;

$\Delta_{electron\ gain} H^\Theta$ for $Cl_{(g)}$ = -348.6 kJ mol^{-1} ;

$\Delta_{bond} H^\Theta$ for Cl_2 = 243.0 kJ mol^{-1}

The magnitude of lattice enthalpy of KCl in kJ mol^{-1} is _____ (Nearest integer)

Open solution

Q21. Numerical

The total number of negative charge in the tetrapeptide, Gly-Glu-Asp-Tyr, at pH 12.5 will be _____. (Integer answer)

Open solution

Q22. Numerical

An aqueous KCl solution of density 1.20 g mL^{-1} has a molality of 3.30 mol kg^{-1}. The molarity of the solution in mol L^{-1} is _____ (Nearest integer) [Molar mass of KCl = 74.5]

Open solution

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Mathematics 20 questions

Q23. MCQ (Single Correct Answer)

The mean and standard deviation of 20 observations were calculated as 10 and 2.5 respectively. It was found that by mistake one data value was taken as 25 instead of 35. If α and $\sqrt{\beta}$ are the mean and standard deviation respectively for correct data, then (α, β) is :

(A) (11, 26) (B) (10.5, 25) (C) (11, 25) (D) (10.5, 26)

Open solution

Q24. MCQ (Single Correct Answer)

Let y = y(x) be a solution curve of the differential equation $(y+1)\tan^2 x\,dx + \tan x\,dy + y\,dx = 0$, $x \in \left(0, \frac{\pi}{2}\right)$. If $\lim\limits_{x\to 0+} xy(x) = 1$, then the value of $y\left(\frac{\pi}{4}\right)$ is :

(A) $-\frac{\pi}{4}$ (B) $\frac{\pi}{4} - 1$ (C) $\frac{\pi}{4} + 1$ (D) $\frac{\pi}{4}$

Open solution

Q25. MCQ (Single Correct Answer)

Let A and B be independent events such that P(A) = p, P(B) = 2p. The largest value of p, for which P (exactly one of A, B occurs) = $\frac{5}{9}$, is :

(A) $\frac{1}{3}$ (B) $\frac{2}{9}$ (C) $\frac{4}{9}$ (D) $\frac{5}{12}$

Open solution

Q26. MCQ (Single Correct Answer)

Let $\theta \in \left(0, \frac{\pi}{2}\right)$. If the system of linear equations

$$(1+\cos^2\theta)x + \sin^2\theta y + 4\sin 3\theta z = 0$$

$$\cos^2\theta x + (1+\sin^2\theta)y + 4\sin 3\theta z = 0$$

$$\cos^2\theta x + \sin^2\theta y + (1+4\sin 3\theta)z = 0$$

has a non-trivial solution, then the value of θ is :

(A) $\frac{4\pi}{9}$ (B) $\frac{7\pi}{18}$ (C) $\frac{\pi}{18}$ (D) $\frac{5\pi}{18}$

Open solution

Q27. MCQ (Single Correct Answer)

Let $f(x) = \cos\left(2\tan^{-1}\sin\left(\cot^{-1}\sqrt{\frac{1-x}{x}}\right)\right)$, 0 < x < 1. Then :

(A) $(1-x)^2 f'(x) - 2(f(x))^2 = 0$ (B) $(1+x)^2 f'(x) + 2(f(x))^2 = 0$

(C) $(1-x)^2 f'(x) + 2(f(x))^2 = 0$ (D) $(1+x)^2 f'(x) - 2(f(x))^2 = 0$

Open solution

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Q28. MCQ (Single Correct Answer)

Out of all the patients in a hospital 89% are found to be suffering from heart ailment and 98% are suffering from lungs infection. If K% of them are suffering from both ailments, then K can not belong to the set :

Open solution

(A) {80, 83, 86, 89} (B) {84, 86, 88, 90} (C) {79, 81, 83, 85} (D) {84, 87, 90, 93}

Q29. MCQ (Single Correct Answer)

The equation $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$ represents a circle with :

Open solution

(A) centre at $(0, -1)$ and radius $\sqrt{2}$ (B) centre at (0, 1) and radius $\sqrt{2}$

(C) centre (0, 0) and radius $\sqrt{2}$ (D) centre at (0, 1) and radius 2

Q30. MCQ (Single Correct Answer)

If a line along a chord of the circle $4x^2 + 4y^2 + 120x + 675 = 0$, passes through the point $(-30, 0)$ and is tangent to the parabola $y^2 = 30x$, then the length of this chord is :

Open solution

(A) 5 (B) 7 (C) $5\sqrt{3}$ (D) $3\sqrt{5}$

Q31. MCQ (Single Correct Answer)

The value of $\int_{\frac{-1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \left(\left(\frac{x+1}{x-1}\right)^2 + \left(\frac{x-1}{x+1}\right)^2 - 2\right)^{\frac{1}{2}} dx$ is :

Open solution

(A) $\log_e 4$ (B) $\log_e 16$

(C) $2\log_e 16$ (D) $4\log_e (3 + 2\sqrt{2})$

Q32. MCQ (Single Correct Answer)

If $A = \begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ i & 1 \end{pmatrix}$, $i = \sqrt{-1}$, and Q = $A^T BA$, then the inverse of the matrix A Q^{2021} A^T is equal to :

Open solution

(A) $\begin{pmatrix} \frac{1}{\sqrt{5}} & -2021 \\ 2021 & \frac{1}{\sqrt{5}} \end{pmatrix}$

(B) $\begin{pmatrix} 1 & 0 \\ -2021i & 1 \end{pmatrix}$

(C) $\begin{pmatrix} 1 & 0 \\ 2021i & 1 \end{pmatrix}$

(D) $\begin{pmatrix} 1 & -2021i \\ 0 & 1 \end{pmatrix}$

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Q33. MCQ (Single Correct Answer)

If the sum of an infinite GP a, ar, ar^2, ar^3, ... is 15 and the sum of the squares of its each term is 150, then the sum of ar^2, ar^4, ar^6, ... is :

(A) $\frac{5}{2}$ (B) $\frac{1}{2}$ (C) $\frac{25}{2}$ (D) $\frac{9}{2}$

Open solution

Q34. Numerical

Let $z = \frac{1-i\sqrt{3}}{2}$, $i = \sqrt{-1}$. Then the value of $21 + \left(z + \frac{1}{z}\right)^3 + \left(z^2 + \frac{1}{z^2}\right)^3 + \left(z^3 + \frac{1}{z^3}\right)^3 + \ldots + \left(z^{21} + \frac{1}{z^{21}}\right)^3$ is _____.

Open solution

Q35. Numerical

The sum of all integral values of k ($k \neq 0$) for which the equation $\frac{2}{x-1} - \frac{1}{x-2} = \frac{2}{k}$ in x has no real roots, is _____.

Open solution

Q36. Numerical

If ${}^1P_1 + 2.{}^2P_2 + 3.{}^3P_3 + \ldots + 15.{}^{15}P_{15} = {}^qP_r - s, 0 \leq s \leq 1$, then ${}^{q+s}C_{r-s}$ is equal to _____.

Open solution

Q37. Numerical

A wire of length 36 m is cut into two pieces, one of the pieces is bent to form a square and the other is bent to form a circle. If the sum of the areas of the two figures is minimum, and the circumference of the circle is k (meter), then $\left(\frac{4}{\pi} + 1\right)k$ is equal to _____.

Open solution

Q38. Numerical

The area of the region $S = \{(x, y) : 3x^2 \leq 4y \leq 6x + 24\}$ is _____.

Open solution

Q39. Numerical

The locus of a point, which moves such that the sum of squares of its distances from the points (0, 0), (1, 0), (0, 1), (1, 1) is 18 units, is a circle of diameter d. Then d^2 is equal to _____.

Open solution

Q40. Numerical

If y = y(x) is an implicit function of x such that $\log_e(x + y) = 4xy$, then $\frac{d^2y}{dx^2}$ at x = 0 is equal to _____.

Open solution

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Q41. Numerical

The number of three-digit even numbers, formed by the digits 0, 1, 3, 4, 6, 7 if the repetition of digits is not allowed, is _____.

Open solution

Q42. Numerical

Let a, b $\in$ R, b $\neq$ 0, Define a function

$$f(x) = \begin{cases} a\sin\frac{\pi}{2}(x-1), & for\, x \leq 0 \\ \frac{\tan 2x - \sin 2x}{bx^3}, & for\, x > 0 \end{cases}.$$

If f is continuous at x = 0, then $10 - ab$ is equal to _____.

Open solution

Physics 28 questions

Q43. MCQ (Single Correct Answer)

The fractional change in the magnetic field intensity at a distance 'r' from centre on the axis of current carrying coil of radius 'a' to the magnetic field intensity at the centre of the same coil is : (Take r < a)

(A) $\frac{3}{2}\frac{a^2}{r^2}$

(B) $\frac{2}{3}\frac{a^2}{r^2}$

(C) $\frac{2}{3}\frac{r^2}{a^2}$

(D) $\frac{3}{2}\frac{r^2}{a^2}$

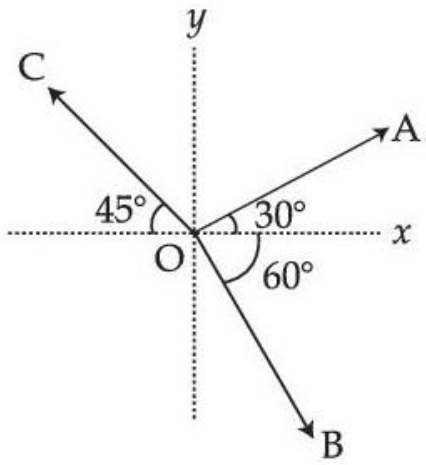
Open solution

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Q44. MCQ (Single Correct Answer)

Open solution

The magnitude of vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ in the given figure are equal. The direction of $\overrightarrow{OA} + \overrightarrow{OB} - \overrightarrow{OC}$ with x-axis will be :



(A) $\tan^{-1}\frac{(1-\sqrt{3}-\sqrt{2})}{(1+\sqrt{3}+\sqrt{2})}$

(B) $\tan^{-1}\frac{(\sqrt{3}-1+\sqrt{2})}{(1+\sqrt{3}-\sqrt{2})}$

(C) $\tan^{-1}\frac{(\sqrt{3}-1+\sqrt{2})}{(1-\sqrt{3}+\sqrt{2})}$

(D) $\tan^{-1}\frac{(1+\sqrt{3}-\sqrt{2})}{(1-\sqrt{3}-\sqrt{2})}$

Q45. MCQ (Single Correct Answer)

Open solution

Car B overtakes another car A at a relative speed of 40 ms^{-1}. How fast will the image of car B appear to move in the mirror of focal length 10 cm fitted in car A, when the car B is 1.9 m away from the car A?

(A) 4 ms^{-1}

(B) 0.2 ms^{-1}

(C) 40 ms^{-1}

(D) 0.1 ms^{-1}

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Q46. MCQ (Single Correct Answer)

Open solution

Inside a uniform spherical shell :

(1) the gravitational field is zero

(2) the gravitational potential is zero

(3) the gravitational field is same everywhere

(4) the gravitational potential is same everywhere

(5) all of the above

Choose the most appropriate answer from the options given below :

(A) (1), (3) and (4) only

(B) (5) only

(C) (1), (2) and (3) only

(D) (2), (3) and (4) only

Q47. MCQ (Single Correct Answer)

Open solution

Two narrow bores of diameter 5.0 mm and 8.0 mm are joined together to form a U-shaped tube open at both ends. If this U-tube contains water, what is the difference in the level of two limbs of the tube. [Take surface tension of water T = 7.3×10^{-2} Nm^{-1}, angle of contact = 0, g = 10 ms^2 and density of water = 1.0×10^3 kg m^{-3}]

(A) 3.62 mm

(B) 2.19 mm

(C) 5.34 mm

(D) 4.97 mm

Q48. MCQ (Single Correct Answer)

Open solution

An electric appliance supplies 6000 J/min heat to the system. If the system delivers a power of 90W. How long it would take to increase the internal energy by 2.5×10^3 J ?

(A) 2.5×10^2 s

(B) 4.1×10^1 s

(C) 2.4×10^3 s

(D) 2.5×10^1 s

Q49. MCQ (Single Correct Answer)

Open solution

An inductor coil stores 64 J of magnetic field energy and dissipates energy at the rate of 640 W when a current of 8A is passed through it. If this coil is joined across an ideal battery, find the time constant of the circuit in seconds :

(A) 0.4

(B) 0.8

(C) 0.125

(D) 0.2

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Q50. MCQ (Single Correct Answer)

A series LCR circuit driven by 300 V at a frequency of 50 Hz contains a resistance R = 3 kΩ, an inductor of inductive reactance X_L = 250 $\pi\Omega$ and an unknown capacitor. The value of capacitance to maximize the average power should be : (Take π^2 = 10)

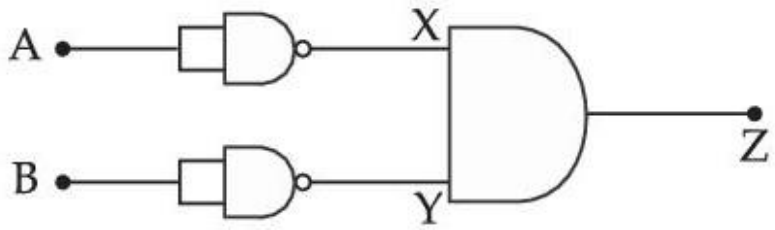
Open solution

(A) 4 μF (B) 25 μF (C) 400 μF (D) 40 μF

Q51. MCQ (Single Correct Answer)

Identify the logic operation carried out by the given circuit :-

Open solution



(A) OR (B) AND (C) NOR (D) NAND

Q52. MCQ (Single Correct Answer)

A particular hydrogen like ion emits radiation of frequency 2.92×10^{15} Hz when it makes transition from n = 3 to n = 1. The frequency in Hz of radiation emitted in transition from n = 2 to n = 1 will be :

Open solution

(A) 0.44×10^{15} (B) 6.57×10^{15} (C) 4.38×10^{15} (D) 2.46×10^{15}

Q53. MCQ (Single Correct Answer)

In a photoelectric experiment ultraviolet light of wavelength 280 nm is used with lithium cathode having work function ϕ = 2.5 eV. If the wavelength of incident light is switched to 400 nm, find out the change in the stopping potential. (h = 6.63×10^{-34} Js, c = 3×10^8 ms^{-1})

Open solution

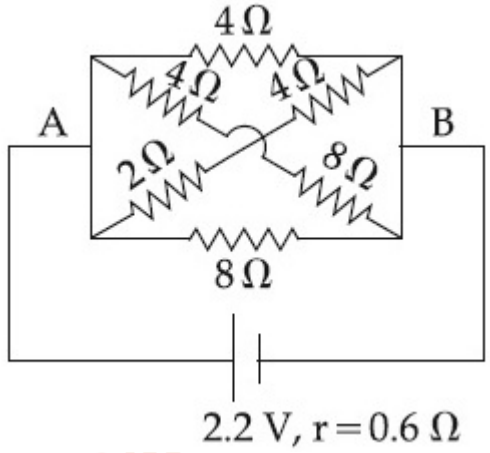
(A) 1.3 V (B) 1.1 V (C) 1.9 V (D) 0.6 V

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Q54. MCQ (Single Correct Answer)

Open solution

In the given figure, the emf of the cell is 2.2 V and if internal resistance is 0.6Ω. Calculate the power dissipated in the whole circuit :



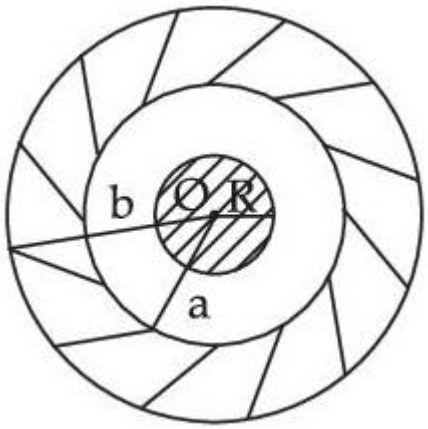
(A) 1.32 W (B) 0.65 W (C) 2.2 W (D) 4.4 W

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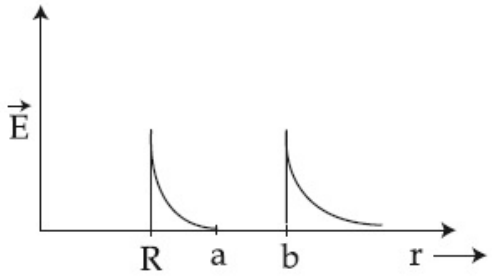
Q55. MCQ (Single Correct Answer)

Open solution

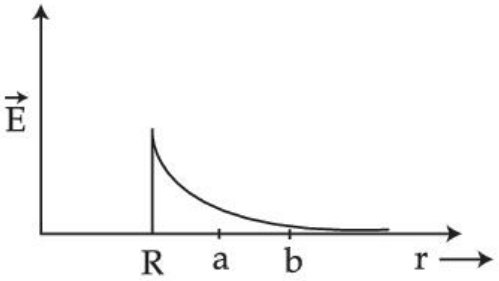
A solid metal sphere of radius R having charge q is enclosed inside the concentric spherical shell of inner radius a and outer radius b as shown in the figure. The approximate variation of electric field $\vec{E}$ as a function of distance r from centre O is given by



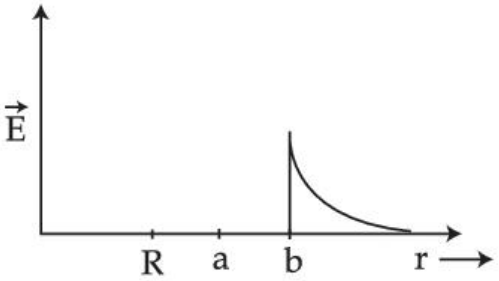
(A)



(B)

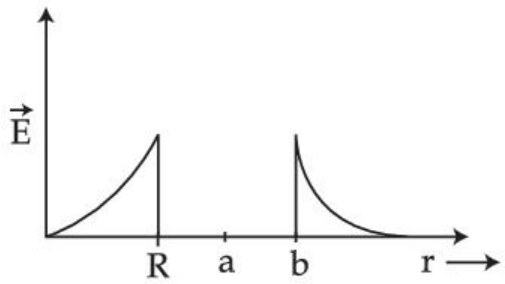


(C)



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(D)



Q56. MCQ (Single Correct Answer)

The rms speeds of the molecules of Hydrogen, Oxygen and Carbon dioxide at the same temperature are V_H, V_O and V_C respectively then :

(A) $V_H > V_O > V_C$ (B) $V_C > V_O > V_H$ (C) $V_H = V_O > V_C$ (D) $V_H = V_O = V_C$

Open solution

Q57. MCQ (Single Correct Answer)

In a Screw Gauge, fifth division of the circular scale coincides with the reference line when the ratchet is closed. There are 50 divisions on the circular scale, and the main scale moves by 0.5 mm on a complete rotation. For a particular observation the reading on the main scale is 5 mm and the 20th division of the circular scale coincides with reference line. Calculate the true reading.

(A) 5.00 mm (B) 5.25 mm (C) 5.15 mm (D) 5.20 mm

Open solution

Q58. MCQ (Single Correct Answer)

What equal length of an iron wire and a copper-nickel alloy wire, each of 2 mm diameter connected parallel to give an equivalent resistance of 3Ω ?

(Given resistivities of iron and copper-nickel alloy wire are 12 $\mu\Omega$ cm and 51 $\mu\Omega$ cm respectively)

(A) 82 m (B) 97 m (C) 110 m (D) 90 m

Open solution

Q59. MCQ (Single Correct Answer)

The initial mass of a rocket is 1000 kg. Calculate at what rate the fuel should be burnt so that the rocket is given an acceleration of 20 ms^{-2}. The gases come out at a relative speed of 500 ms^{-1} with respect to the rocket : [Use g = 10 m/s^2]

(A) 6.0×10^2 kg s^{-1} (B) 500 kg s^{-1}

(C) 10 kg s^{-1} (D) 60 kg s^{-1}

Open solution

Q60. MCQ (Single Correct Answer)

If E, L, M and G denote the quantities as energy, angular momentum, mass and constant of gravitation respectively, then the dimensions of P in the formula $P = EL^2M^{-5}G^{-2}$ are :

(A) $[M^0 L^1 T^0]$ (B) $[M^{-1} L^{-1} T^2]$ (C) $[M^1 L^1 T^{-2}]$ (D) $[M^0 L^0 T^0]$

Open solution

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Q61. MCQ (Single Correct Answer)

The material filled between the plates of a parallel plate capacitor has resistivity 200 Ωm. The value of capacitance of the capacitor is 2 pF. If a potential difference of 40 V is applied across the plates of the capacitor, then the value of leakage current flowing out of the capacitor is : (given the value of relative permittivity of material is 50)

(A) 9.0 μA (B) 9.0 mA (C) 0.9 mA (D) 0.9 μA

Open solution

Q62. MCQ (Single Correct Answer)

Statement I : By doping silicon semiconductor with pentavalent material, the electrons density increases.

Statement II : The n-type semiconductor has net negative charge.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement - I is true but Statement - II is false.

(B) Statement - I is false but Statement - II is true.

(C) Both Statement I and Statement II are true.

(D) Both Statement I and Statement II are false.

Open solution

Q63. Numerical

A uniform chain of length 3 meter and mass 3 kg overhangs a smooth table with 2 meter lying on the table. If k is the kinetic energy of the chain in joule as it completely slips off the table, then the value of k is ..... . (Take g = 10 m/s^2)

Open solution

Q64. Numerical

The electric field in a plane electromagnetic wave is given by

$\overrightarrow{E} = 200\cos\left[\left(\frac{0.5\times10^3}{m}\right)x - \left(1.5\times10^{11}\,\frac{rad}{s}\times t\right)\right]\frac{V}{m}\hat{j}$. If this wave falls normally on a perfectly reflecting surface having an area of 100 cm^2. If the radiation pressure exerted by the E.M. wave on the surface during a 10 minute exposure is $\frac{x}{10^9}\frac{N}{m^2}$. Find the value of x .

Open solution

Q65. Numerical

Two spherical balls having equal masses with radius of 5 cm each are thrown upwards along the same vertical direction at an interval of 3s with the same initial velocity of 35 m/s, then these balls collide at a height of ..... m. (Take g = 10 m/s^2)

Open solution

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Q66. Numerical

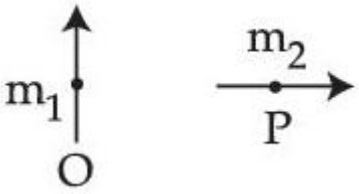
A soap bubble of radius 3 cm is formed inside the another soap bubble of radius 6 cm. The radius of an equivalent soap bubble which has the same excess pressure as inside the smaller bubble with respect to the atmospheric pressure is ..... cm.

Open solution

Q67. Numerical

Two short magnetic dipoles m_1 and m_2 each having magnetic moment of 1 Am^2 are placed at point O and P respectively. The distance between OP is 1 meter. The torque experienced by the magnetic dipole m_2 due to the presence of m_1 is ..... $\times 10^{-7}$ Nm.

Open solution



Q68. Numerical

Two travelling waves produce a standing wave represented by equation,

$y = 1.0\ mm\ \cos(1.57\ cm^{-1})\ x\ \sin(78.5\ s^{-1})t.$

The node closest to the origin in the region $x > 0$ will be at x = ... cm.

Open solution

Q69. Numerical

White light is passed through a double slit and interference is observed on a screen 1.5 m away. The separation between the slits is 0.3 mm. The first violet and red fringes are formed 2.0 mm and 3.5 mm away from the central white fringes. the difference in wavelengths of red and violet light is ..... nm.

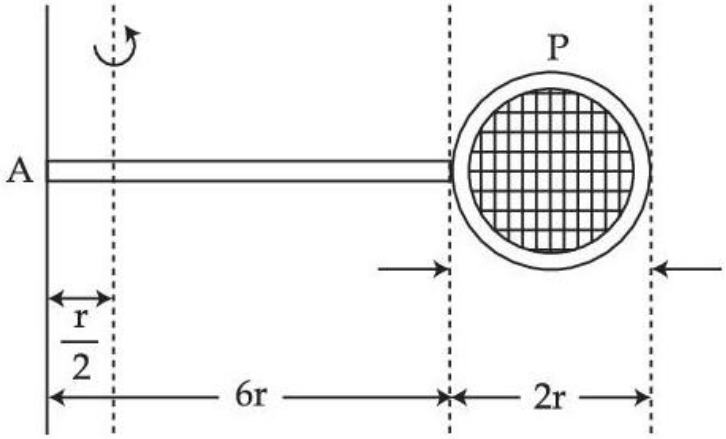
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Q70. Numerical

Consider a badminton racket with length scales as shown in the figure.

Open solution



If the mass of the linear and circular portions of the badminton racket are same (M) and the mass of the threads are negligible, the moment of inertia of the racket about an axis perpendicular to the handle and in the plane of the ring at $\frac{r}{2}$ distance from the end A of the handle will be ..... Mr^2.

Out of syllabus 20 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q71. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following is correct for the adsorption of a gas at a given temperature on a solid surface?

Open solution

(A) $\Delta H > 0, \Delta S > 0$

(B) $\Delta H > 0, \Delta S < 0$

(C) $\Delta H < 0, \Delta S < 0$

(D) $\Delta H < 0, \Delta S > 0$

Q72. MCQ (Single Correct Answer) · **Chemistry**

The polymer formed on heating Novolac with formaldehyde is :

Open solution

(A) Bakelite

(B) Polyester

(C) Melamine

(D) Nylon 6, 6

Q73. MCQ (Single Correct Answer) · **Chemistry**

The conversion of hydroxyapatite occurs due to presence of F^- ions in water. The correct formula of hydroxyapatite is :

Open solution

(A) $[3Ca_3(PO_4)_2 . Ca(OH)_2]$

(B) $[3Ca(OH)_2 . CaF_2]$

(C) $[Ca_3(PO_4)_2 . CaF_2]$

(D) $[3Ca_3(PO_4)_2 . CaF_2]$

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Q74. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following methods is most suitable for preparing deionized water?

(A) Synthetic resin method
(B) Clark's method
(C) Calgon's method
(D) Permutit method

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements :

Statement I : The choice of reducing agents for metals extraction can be made by using Ellingham diagram, a plot of ΔG vs temperature.

Statement II : The value of ΔS increases from left to right in Ellingham diagram.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are true
(B) Statement I is false but Statement II is true
(C) Both Statement I and Statement II are false
(D) Statement I is true but Statement II is false

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

What are the products formed in sequence when excess of CO_2 is passed in slaked lime?

(A) $Ca(HCO_3)_2$, $CaCO_3$
(B) $CaCO_3$, $Ca(HCO_3)_2$
(C) CaO, $Ca(HCO_3)_2$
(D) CaO, $CaCO_3$

Open solution

Q77. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements.

Statement I : Frenkel defects are vacancy as well as interstitial defects.

Statement II : Frenkel defect leads to colour in ionic solids due to presence of F-centres.

Choose the most appropriate answer for the statements from the options given below :

(A) Statement I is false but Statement II is true
(B) Both Statement I and Statement II are true
(C) Statement I is true but Statement II is false
(D) Both Statement I and Statement II are false

Open solution

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Q78. MCQ (Single Correct Answer) · **Chemistry**

The incorrect statement is :

(A) Cl_2 is more reactive than ClF.

(B) F_2 is more reactive than ClF.

(C) On hydrolysis ClF forms HOCl and HF.

(D) F_2 is a stronger oxidizing agent than Cl_2 in aqueous solution.

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

The sum of solutions of the equation

$\frac{\cos x}{1+\sin x} = |\tan 2x|, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) - \left\{\frac{\pi}{4}, -\frac{\pi}{4}\right\}$ is :

(A) $-\frac{11\pi}{30}$ (B) $\frac{\pi}{10}$ (C) $-\frac{7\pi}{30}$ (D) $-\frac{\pi}{15}$

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

On the ellipse $\frac{x^2}{8} + \frac{y^2}{4} = 1$ let P be a point in the second quadrant such that the tangent at P to the ellipse is perpendicular to the line x + 2y = 0. Let S and S' be the foci of the ellipse and e be its eccentricity. If A is the area of the triangle SPS' then, the value of $(5 - e^2)$. A is :

(A) 6 (B) 12 (C) 14 (D) 24

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

The sum of the series

$\frac{1}{x+1} + \frac{2}{x^2+1} + \frac{2^2}{x^4+1} + \ldots\ldots + \frac{2^{100}}{x^{2^{100}}+1}$ when x = 2 is :

(A) $1 + \frac{2^{101}}{4^{101}-1}$ (B) $1 + \frac{2^{100}}{4^{101}-1}$

(C) $1 - \frac{2^{100}}{4^{100}-1}$ (D) $1 - \frac{2^{101}}{2^{400}-1}$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

If ${}^{20}C_r$ is the co-efficient of x^r in the expansion of $(1 + x)^{20}$, then the value of $\sum_{r=0}^{20} r^2 . {}^{20}C_r$ is equal to :

(A) 420×2^{19} (B) 380×2^{19} (C) 380×2^{18} (D) 420×2^{18}

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{j} - \hat{k}$. If $\vec{c}$ is a vector such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a}.\vec{c} = 3$, then $\vec{a}.(\vec{b} \times \vec{c})$ is equal to :

(A) -2 (B) -6 (C) 6 (D) 2

Open solution

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Q84. MCQ (Single Correct Answer) · **Mathematics**

A plane P contains the line $x + 2y + 3z + 1 = 0 = x - y - z - 6$, and is perpendicular to the plane $-2x + y + z + 8 = 0$. Then which of the following points lies on P?

(A) $(-1, 1, 2)$ (B) $(0, 1, 1)$ (C) $(1, 0, 1)$ (D) $(2, -1, 1)$

Open solution

Q85. MCQ (Single Correct Answer) · **Mathematics**

The value of

$\lim_{n\to\infty} \frac{1}{n} \sum_{r=0}^{2n-1} \frac{n^2}{n^2+4r^2}$ is :

(A) $\frac{1}{2}\tan^{-1}(2)$ (B) $\frac{1}{2}\tan^{-1}(4)$ (C) $\tan^{-1}(4)$ (D) $\frac{1}{4}\tan^{-1}(4)$

Open solution

Q86. MCQ (Single Correct Answer) · **Mathematics**

Let ABC be a triangle with A$(-3, 1)$ and $\angle ACB = \theta$, $0 < \theta < \frac{\pi}{2}$. If the equation of the median through B is $2x + y - 3 = 0$ and the equation of angle bisector of C is $7x - 4y - 1 = 0$, then $\tan\theta$ is equal to :

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) $\frac{4}{3}$ (D) 2

Open solution

Q87. MCQ (Single Correct Answer) · **Mathematics**

If the truth value of the Boolean expression $((p \vee q) \wedge (q \to r) \wedge (\sim r)) \to (p \wedge q)$ is false, then the truth values of the statements p, q, r respectively can be :

(A) T F T (B) F F T (C) T F F (D) F T F

Open solution

Q88. Numerical · **Mathematics**

Let the line L be the projection of the line $\frac{x-1}{2} = \frac{y-3}{1} = \frac{z-4}{2}$ in the plane $x - 2y - z = 3$. If d is the distance of the point (0, 0, 6) from L, then d^2 is equal to _____.

Open solution

Q89. Numerical · **Physics**

A source and a detector move away from each other in absence of wind with a speed of 20 m/s with respect to the ground. If the detector detects a frequency of 1800 Hz of the sound coming from the source, then the original frequency of source considering speed of sound in air 340 m/s will be ..... Hz.

Open solution

Q90. Numerical · **Physics**

An amplitude modulated wave is represented by
$C_m(t) = 10(1 + 0.2 \cos 12560t) \sin(111 \times 10^4 t)$ volts. The modulating frequency in kHz will be ..... .

Open solution

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Answer Key

Chemistry

1. **D**	2. **A**	3. **A**	4. **C**	5. **A**
6. **D**	7. **B**	8. **B**	9. **B**	10. **C**
11. **D**	12. **D**	13. **2**	14. **3**	15. **4**
16. **3**	17. **7**	18. **5**	19. **1**	20. **718**
21. **4**	22. **3**			

Mathematics

23. **D**	24. **D**	25. **D**	26. **B**	27. **C**
28. **C**	29. **B**	30. **D**	31. **B**	32. **B**
33. **B**	34. **13**	35. **66**	36. **136**	37. **36**
38. **27**	39. **16**	40. **40**	41. **52**	42. **14**

Physics

43. **D**	44. **A**	45. **D**	46. **A**	47. **B**
48. **A**	49. **D**	50. **A**	51. **C**	52. **D**
53. **A**	54. **C**	55. **A**	56. **A**	57. **C**
58. **B**	59. **D**	60. **D**	61. **C**	62. **A**
63. **40**	64. **354**	65. **50**	66. **2**	67. **1**
68. **1**	69. **300**	70. **52**		

Out of syllabus

71. **C**	72. **A**	73. **A**	74. **A**	75. **D**
76. **B**	77. **C**	78. **A**	79. **A**	80. **A**
81. **D**	82. **D**	83. **A**	84. **B**	85. **B**
86. **C**	87. **C**	88. **26**	89. **2025**	90. **2**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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